Beyond Retrieval: Learning the Applicability Boundaries of Agent Memory
FRONTIER: Decision-Directed Probes and Representation Revision for Frozen Coding Agents

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Status: no Gemma run, software-issue benchmark, or competition-harness integration is reported.

Abstract

A correction verified on one task can become harmful when reused on a superficially similar task. We propose FRONTIER, a frozen-backbone coding-agent architecture that treats memory applicability as an unresolved decision problem rather than a similarity threshold. A memory record stores competing applicability hypotheses, executable observations, counterexamples, and a versioned reuse condition. Before reuse, the controller either finds an action optimal under every retained hypothesis or selects a sandboxed probe intended to reduce decision uncertainty. Outcomes that contradict the entire hypothesis set trigger representation revision rather than automatic confidence escalation. We formalize the distinction between identifying a mechanism and identifying a sufficient action, establish conditional limitations and stopping results, and supply an exactly enumerated four-hypothesis example. In that constructed example, an entropy-maximizing probe leaves decision error unchanged, while a lower-information probe resolves the action using the same one-probe budget. An additional out-of-class case defeats the apparent certificate and demonstrates why hypothesis coverage cannot be assumed. The contribution is an agent-memory architecture organized around decision-directed boundary discovery, not a new principle of Bayesian experimental design. Actual gains for Gemma coding agents remain unmeasured.

1. Introduction

Memory can fail without containing a false historical statement. A stored correction may have passed its original checks yet require a condition absent from the current task. Retrieval relevance and source-task verification do not establish current applicability.

For example, replacing > with >= can repair an inclusive boundary requirement and break a strict one. A retrieved patch is not sufficient evidence that the current requirement is inclusive. The agent may need a discriminating observation before using it.

FRONTIER asks: what is the least investigation needed to decide whether and how a memory should influence the next action? Its aim is not to eliminate uncertainty about an entire repository. It is to resolve uncertainty that changes the decision.

We contribute: (i) a typed architecture separating repository facts from applicability hypotheses; (ii) an executable decision rule with explicit unresolved and frame-failure states; (iii) conditional propositions and a reproducible separation example; and (iv) a proposed evaluation isolating probe selection, stored boundaries, and representation revision. We do not claim first invention of active diagnosis, conditional memory, version spaces, or decision-theoretic experiment selection.

2. Related work and scope of the claim

RepoGraph supplies repository-level code structure to software-engineering agents. FRONTIER can use such structure to propose investigations, but a call edge is not evidence that a correction applies. See Ouyang et al., *RepoGraph: Enhancing AI Software Engineering with Repository-level Code Graph* (2024), https://arxiv.org/abs/2410.14684.

Reflexion retains verbal feedback without updating backbone weights. Freezing the backbone while changing external memory is therefore not our novelty. See Shinn et al., *Reflexion: Language Agents with Verbal Reinforcement Learning* (2023), https://proceedings.neurips.cc/paper_files/paper/2023/hash/1b44b878bb782e6954cd888628510e90-Abstract-Conference.html.

CounterMem checks alternative actions, stores corrections with applicability conditions, and trains a selector that can skip retrieved memory. It is a particularly close comparator. FRONTIER's proposed distinction is online acquisition of observations specifically chosen to resolve disagreement about reuse, with explicit revision of the applicability representation when observations invalidate all candidates. This distinction is an architectural proposal, not an established empirical advantage. See Pu, Tang, and Zhang, *CounterMem: World-Model-Verified Counterfactual Memory for Language Agents* (2026), https://arxiv.org/abs/2609.31874.

InspectCoder already performs interactive dynamic investigation and runtime perturbation for program repair. FRONTIER is not the first agent that experiments. Its proposed memory object is the reusable boundary and its discriminating probe, not only a current-task diagnosis. See Wang et al., *InspectCoder* (2025), https://arxiv.org/abs/2510.18327.

Choosing experiments for a downstream decision is classical Bayesian decision theory, not a new information principle; see Chaloner and Verdinelli, *Bayesian Experimental Design: A Review* (1995), https://doi.org/10.1214/ss/1177009939. LLM-driven hypothesis-family revision and experiment design are also studied in *Bayesian Model Discovery with Language Models: Closing the Loop between Hypotheses and Experiments* (2026), https://arxiv.org/abs/2610.25747. FRONTIER specializes these ideas to contested memory reuse and makes their assumptions visible. A comprehensive novelty audit remains necessary.

3. Architecture

3.1 Memory and two typed graphs

A memory record is

m = (r, H, E_positive, E_negative, g, version).

Here r is a reusable action schema; H is a finite set of competing applicability mechanisms; E_positive and E_negative are executed observations and counterexamples; g is a proposed applicability condition; and version identifies its representation, provenance, and environment assumptions.

The repository graph stores code facts such as imports, calls, definitions, and observed execution. The epistemic graph stores predicts, contradicts, distinguishes, and conditionally-authorizes relations. These edge types are not interchangeable. Neither shared neighbors nor semantic similarity constitutes a correctness certificate.

A hypothesis must predict observable outcomes for specified probes and a loss or acceptable-action set for the current decision. Natural-language guesses without these commitments remain unscored proposals. A generated hypothesis is not accepted as a fact because the backbone finds it plausible.

3.2 The controller's states

Each attempted reuse is in one of four states:

1. PROPOSED: retrieval supplied a potentially relevant record.
2. DECISION-RESOLVED: retained hypotheses agree on an acceptable action.
3. UNRESOLVED: available evidence does not resolve the action.
4. FRAME-FAILURE: no retained hypothesis explains an executed observation.

DECISION-RESOLVED is explicitly model-relative. It is not a guarantee that the true mechanism is represented. An unmodeled alternative is always a deployment concern; blindly adding an unrestricted unknown hypothesis would instead prevent every nontrivial certificate.

3.3 Representation and expansion

The initial frame is a restricted language of predicates over observable types, boundary requirements, configuration, dependency versions, and runtime traces. Candidate mechanisms combine these predicates with testable predictions.

On frame failure, permitted expansion operators add an observable predicate, split a category, introduce a relation, or revise an environment dependency. A new candidate must account for the recorded contradiction and predict a fresh discriminating observation. Replaying old evidence alone is not validation. Old conditions are retained with their original frame version; they are not silently upgraded to universal rules.

The expansion policy is proposed, not implemented by the accompanying finite example. A real implementation must specify its language, operator ordering, candidate cap, and execution limits before evaluation.

4. Decision-directed investigation

Let H be the retained hypotheses, A the candidate actions, Q the legal probes, and L(h,a) the predicted loss. Each probe q has cost c(q)>0 and possible observation o. In the deterministic analysis,

H_after(q,o) = {h in H : h(q)=o}.

Unknown predictions are not silently treated as contradictions. They require an unresolved status or a larger model; the proofs below assume all relevant predictions are specified.

Define the common-optimum set

C(H) = intersection over h in H of argmin over a in A L(h,a).

If C(H) is nonempty, the controller can act without identifying one unique mechanism. Otherwise it may probe. With a positive prior p(h), Bayesian decision risk is

B(H) = min_a sum_h p(h | E) L(h,a).

The one-step decision value of q is

V_dec(q) = B(H) - sum_o p(o | E,q) B(H_after(q,o)).

A practical policy can rank probes by V_dec(q)/c(q), accounting for candidate generation and execution in its budget. This is a myopic policy, not a claim of globally optimal diagnosis. It may prefer a low-information observation when that observation settles the action. Multi-step planning, noisy results, and prior calibration are separate engineering problems.

Algorithm: bounded applicability inquiry

Retrieve a memory proposal and instantiate candidate hypotheses.
While budget remains:
If H is empty, enter FRAME-FAILURE and attempt bounded expansion.
If a hypothesis cannot supply required predictions, mark unresolved.
If C(H) is nonempty, return a conditional decision certificate.
Rank legal probes by estimated decision value per cost.
If none is useful or affordable, return UNRESOLVED.
Execute the selected probe in an isolated, resettable state.
Record provenance, update H, and retain counterexamples.
Return UNRESOLVED.

Returning UNRESOLVED does not require abandoning the software issue: the base agent may continue without memory guidance. It must not present the rejected memory as verified. The agent's ordinary final patch evaluation remains necessary.

5. Conditional results

Proposition 1: passive indistinguishability

Suppose two possible situations yield identical passive observations but have disjoint sets of correct actions. No deterministic policy using only those observations can be correct in both. Under equal probability and two mutually exclusive correct actions, even a randomized policy has error at least one half.

Proof. The policy receives the same input in both situations. A fixed decision fails in at least one. If it chooses the first action with probability p, the two equally weighted errors average to one half. Additional information or abstention is necessary. This does not establish that every task requires an active probe.

Proposition 2: action sufficiency without identification

Assume finite A and H, correct loss predictions, and a true hypothesis h_star in H. Every action in C(H) is optimal under h_star, even when multiple mechanisms remain.

Proof. Membership in the intersection implies membership in the optimal-action set of every retained hypothesis, including h_star. The result fails without hypothesis coverage; agreement among wrong hypotheses is not a safety guarantee.

Proposition 3: conditional finite identification bound

If the true deterministic hypothesis remains in H and every selected probe reduces the retained set to at most half its preceding size, at most ceil(log2 |H_initial|) probes identify it.

Proof. After k probes at most |H_initial|/2^k hypotheses remain. Coverage prevents the set becoming empty. Balanced probes need not exist, and complete identification is unnecessary when C(H) becomes nonempty earlier. This is a standard version-space argument, not a claimed new sample-complexity theorem.

6. Executable finite experiment

We exactly enumerate four equally likely hypotheses and two unit-cost probes. A and B denote alternative patch actions; the probe names are mnemonic labels, not tests extracted from an actual repository.

Hypothesis	Signature probe	Boundary probe	Correct action
h0	0	0	A
h1	0	0	A
h2	1	0	A
h3	1	1	B

The signature probe splits H into two equally sized groups, giving 1 bit of information. The boundary probe yields groups of sizes three and one, giving approximately 0.8113 bits. Entropy-greedy selection therefore chooses signature.

However, signature leaves expected Bayes decision error at 0.25: outcome 1 leaves h2 and h3, which demand different actions. Boundary reduces it to zero. After outcome 0, three hypotheses remain, but all authorize A. Identifying the mechanism would add no decision value.

Policy	Correct decisions	Probes per case
Passive majority action	3/4	0
Entropy-greedy, one probe	3/4	1
Decision-directed, one probe	4/4	1

These are exact counts for a constructed example, not measured gains on coding tasks. The example exhibits an objective mismatch under a one-probe budget; it does not show entropy-based investigation is generally inferior.

Misspecification counterexample

Add h4, absent from the controller's initial H, with the same outputs as h2 for both probes but requiring B. Frontier observes boundary=0, conditionally authorizes A, and is wrong. A subsequent independent action check can expose this failure: all remaining hypotheses predicted that A would pass. The controller then enters FRAME-FAILURE. The artifact does not invent h4 or demonstrate automatic recovery.

The standard-library script checks all four in-class cases, this out-of-class witness, and the common-action property on all 15 nonempty subsets of H. No LLM or external dataset is used. The finite construction and both priors and probe tables are fully disclosed; no statistical generalization or hidden benchmark claim is made.

7. Proposed Gemma study

Use one permitted frozen Gemma model across agents and subagents. The backbone proposes candidate mechanisms and probes; a structured controller records predictions and executed outcomes. Use only exposed repository information and public tests, never hidden validation tests. Reset or isolate experimental edits, and account for setup, tokens, hypothesis generation, test execution, and rollback under the task budget. Harness integration is not yet implemented or validated.

Compare: no persistent guidance; retrieved verified corrections; a strong applicability selector; the same probe generator with entropy-based selection; and full decision-directed FRONTIER. Hold the backbone, candidate budget, legal tools, and total execution budget constant. Use faithful existing implementations where available rather than weak approximations. If CounterMem code is unavailable, label a reimplementation explicitly and disclose deviations.

Freeze development choices before a distinct test set. Measure issue resolution, harmful memory reuse, useful abstention from memory, probe costs, and hypothesis-set failure. Ablate stored counterexamples, active probes, and representation expansion separately. Include look-alike tasks where a formerly successful correction becomes harmful, as well as tasks where investigation is unnecessary.

A primary claim requires a predeclared comparator, gain threshold, and uncertainty procedure on independent issues. Matching an established selector at equal budget would not support a claim of superior empirical utility. Extra probes that improve validity but consume the solving budget can still reduce issue resolution. No empirical target is asserted in this submission.

8. Limitations, disclosure, and conclusion

Hypothesis coverage, realistic loss prediction, probe synthesis, safe reset, and representation revision remain unsolved implementation obligations. Unanimity is not truth, priors can be wrong, and incomplete tests can approve an incorrect patch. External memory and executions have costs even when backbone weights are fixed. The graph is an interface for provenance and decisions, not quantum entanglement or an explanation of human discovery.

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FRONTIER's central proposal is to store not only a correction but the inquiry needed to determine when it applies. The finite analysis establishes why passive reuse can be insufficient, why complete identification can be unnecessary, and why an apparently valid decision can fail outside the represented hypothesis class. Whether that architecture benefits small autonomous coding agents is a testable research question, not a reported result.